Experimental Investigations of Water Extraction Process within Permanently Shadowed Regions of the Moon

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ABSTRACT

Heat and mass transfers occurring on the surface and within the subsurface of extraterrestrial bodies during natural and future industrial heating processes require knowledge of thermal properties of porous dry and icy materials. A series of experiments have been employed to simulate borehole heating of icy regolith samples under low vacuum (0.1 to 1.5 Torr) and very low temperatures (>93 K), using JSC-1A analog prepared as “mud-pies” with water contents from 0 to 20% wt. Real-time measurements of temperatures in an array of points and vacuum chamber pressures were recorded. To decipher thermal properties data in this inverse heat equation problem, a 3D finite element method modelling has been used to simulate the behaviour of both the samples and the vacuum chamber. The same model has been used to determine the validity of thermal properties of Lunar icy regolith under extremely high vacuum. The determined thermal conductivities vary with temperature, pressure, ice content, and porosity, while heat capacities vary with temperature, ice content, and porosity. These results should be applicable to research on properties and composition of other rocky and icy extraterrestrial objects or design of water ice prospecting instruments. Most importantly, they should improve our understanding of processes applied in extraterrestrial water mining and visualize the engineering challenges of these processes.

INTRODUCTION

The current paradigm in research of rocky planets, moons and small objects within the Solar System is to “follow the water” – with both institutional and private entities directly interested in this subject due to scientific and commercial reasons. Our understanding of water and other volatiles – their inventory, processes and related phenomena – directly influences knowledge of the evolution of the Solar System and composition of extraterrestrial objects but also it reiterates our plans for space resources utilization. Of all material resources in Space, water has the most promising and possibly fastest commercial utilization potential (Sowers, 2016), (Kornuta, et al., 2019).

On airless or thin-atmosphere bodies between the Sun and the frost line, water exists in the form of ice on the surface and within the subsurface of extraterrestrial objects. These deposits may have various forms – from thick crater deposits as potentially on the Moon and Mercury (Rubanenko, et al., 2019), shallow ice-cemented surface on Mars (Schroghoffer & Forget, 2012) or the Moon (Rubanenko & Aharonson, 2017), to thick deep ice deposits on Mars (Stuurman, et al., 2016).

Regardless of the origin, thermal properties of water ice deposits in planetary environments are not well-constraint and known – this state of being results from dynamic and complex nature of water ice phase change within the porous space. The closest analog for this problem is

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behaviour of permafrost on Earth, for which several modellings and research has been done. However, that research is not directly applicable to this problem beyond Earth due to sublimation-based phase change conditions. Another applicable field of studies is research of freeze-drying processes (or lyophilization) – sublimation-based drying of materials, which often have water and water-ice in their porous space.

Water-ice is known to be more stable in vacuum and atmospheric pressure, when it holds impurities, such as organic (Kossacki, 2019) or inorganic (Kossacki & Leliwa-Kopystynski, 2014) material, or when pure ice is buried (Chevier, et al., 2008), (Kossacki, et al., 2019). That said, mixture of water-ice and other materials behaves differently in the domain of heat and mass transfer than pure water-ice in the same environment.

Thermal properties of dry planetary materials have been measured in situ, through remote sensing and on planetary analogs multiple times. The conclusions of those observations and modellings are primarily that those materials have low or even extremely low thermal conductivities due to conductive and radiative heat transfer modes, which have to rely on small physical contacts between grains of those materials as well as cope with large amounts of empty spaces coming from high porosities. Planetary materials' thermal conductivities are strongly pressure-dependent – addition of gas to the system diametrically shifts main modes of heat transfer to gas conductance when Knudsen number is lower than 10 (Reiss, 2018a).

Concerning icy planetary materials, only a handful of papers that experimentally investigate their thermal properties has been published – one with a scope of Martian icy regolith faithfully recreated through a laborious process of vapour deposition (Siegler, et al., 2012), and one with a scope of Lunar regolith with the extension for icy regolith with water ice up to 1% wt (Reiss, 2018b). There are however simulation studies analysing the heat transfer process in icy deposits based on pre-defined thermal properties (e.g. (Fanale, et al., 1990), (Gori & Corasaniti, 2004), (Hecht & Vasavada, 2006), (Wasilewski, 2018)).

This paper investigates both thermal conductivities and heat capacities of icy JSC-1A materials of ice contents from 0 to 20% wt based on experimental measurements of temperature change in 3 different points of samples during heating under vacuum. Properties of those materials are determined and fitted through a recreation of the experimental situation using COMSOL Multiphysics 5.2 software at the license of the Polish Space Agency.

EXPERIMENTAL METHOD AND SETUP

A series of experiments had been conducted using the facilities of the Center for Space Resources at the Colorado School of Mines. The experiments involved use of a medium vacuum chamber (total measured volume 0.3375 m^3), which is the base for the In Situ Experimental Probe (IEP), an instrument designed to support studies of dry and icy regolith physical properties (Dreyer, et al., 2018).

The vacuum chamber is a two-piece cylindrical structure, with one end fixed with the ground and one end movable as seen in Figure 1. The chamber inner diameter is 45 cm and (when connected) the inner length is 163 cm. The chamber is capable of providing Ultra High Vacuum, however during the course of experiments, only a scroll pump has been employed, which provided vacuum of 0.1 Torr in up to 500 seconds from laboratory ambient pressure (which was lower than standard due to high elevation and oscillated from 600 to 650 Torr).

In all experiments, the same type of regolith analog has been used – JSC 1A. This dry regolith has been tested via optical method to determine its Particle Size Distribution (PSD), which showed that it deviates in comparison to original JSC-1A towards finer particles with

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average particle size by mass (D50) around 400 micrometres, and average particle size by number of particles (Dn50) around 40 micrometres. Full PSD measurement can be seen in Figure 2. Each dry regolith batch used in these experiments before using and testing had been baked in temperature of 200 ^{0}F (93 ^{0}C) for 24 hours.

Figure 1. The vacuum chamber used in this study. CAD model (left, (Dreyer, et al., 2018), with permissions) and author's photo (right).

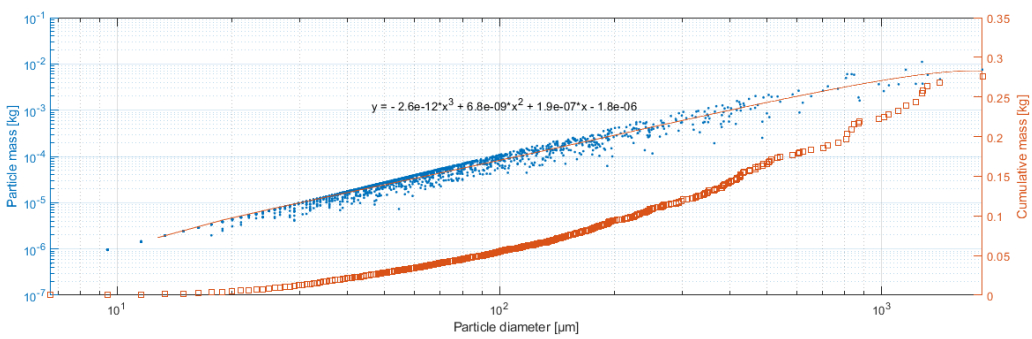

Figure 2. Particle Size Distribution of JSC-1A analog used in this research. Dots represent each particle mass (with line showing fitted trendline), while squares show cumulative mass of the tested batch. Total mass of 6200 particles was 0.2865 kg.

Each experiment consisted of:

- mud-pie batch preparation with specific regolith-water proportions
- cylindrical samples preparation (~500 g, 2.5 in height, 3.28 in diameter)
- samples freezing to -30 ^{0}C
- cryo-cooling of samples in a liquid nitrogen bath
- heater and thermocouples insertion
- sample heating in simulated PSR conditions in the vacuum chamber

At the end of each cryo-cooling, three holes were drilled in the icy regolith samples – one for OMEGA CSH-1011001 cartridge heater with attached thermocouple to be placed in the middle of a sample, one for a thermocouple to be placed in half radius of a sample (around 2 cm or 0.8

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inches from the central hole), and one for a thermocouple at the side of the sample (around 4 cm or 1.6 inches from the central hole). Drilling was not necessary for samples of water content lower than around 2% wt, while richer samples were rock-solid.

The pump-down lasted till pressure reached a plateau, which typically occurred between 0.1 to 0.2 Torr. At the plateau, the heater was turned on and the scroll pump was turned off. The method did not use constant pressure pump-down to enable pressure change monitoring and Chamber Dynamics Analysis (Gertsch, et al., 2017).

Temperature and pressure data had been recorded and the heater was controlled with a dedicated LabView 15 program. Each test run typically lasted around 6 hours. The heater had a DC supply of 32 V and 0.2 A, which translates to around 7 W of power at the heater. However, the current was turned off every time the heater thermocouple read 200 ^{0}C and above. That said, heat flux was not constant.

DATA PROCESSING

This research tried to understand the experimental situation by recreating it through FEM simulations and validating and iterating its dependent parameters to obtain the best fit for heater temperature within first hundreds of seconds of laboratory measurements.

A 3D model of the chamber interior during experiment has been created to simulate study boundaries faithfully. Like in experiments, the modelled chamber provides additional "environmental" heating of the studied samples, magnitude of which had been determined based on experiments without the heater running.

Heat transfer

The heat transfer process in this study is governed by a three-dimensional heat equation accounting for phase change:

$$\rho C_p \frac{\partial T}{\partial t} + \rho C_p \mathbf{u} \cdot \nabla \mathrm{T} + \nabla \cdot \boldsymbol{q} = Q$$

where heat flux from temperature gradient is:

$$\boldsymbol{q} = -k \nabla T$$

while the velocity field comes from the Darcy's law described in a section below.

The physical properties of phase change material are the following:

$$\rho = \theta \rho_{icy} + (1-\theta)\rho_{desiccated}$$
$$C_p = \frac{\theta \rho_{icy} C_{p,icy} + (1-\theta)\rho_{dessicated} C_{p,desiccated}}{\rho} + L \frac{\partial \alpha_m}{\partial T}$$
$$k = \theta k_{icy} + (1-\theta) k_{desiccated}$$
$$\alpha_m = \frac{1}{2} \frac{(1-\theta)\rho_{desiccated} - \theta \rho_{icy}}{\theta \rho_{icy} + (1-\theta)\rho_{desiccated}}$$
$$L = L_s C_i \theta$$

As seen above, the phase change material has been divided into two bulk materials – icy regolith (phase 1, icy), which is composed of mixed properties of both water ice and dry regolith, and desiccated regolith (phase 2, desiccated), which is composed of properties of dry regolith with higher porosity than initial due to full ice desiccation. During calculations, both of those

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materials are seen as bulk and are mixed geometrically through parameter θ, which signals icy phase fraction (with values 0 describing desiccated, 1 describing icy and values in between as transition between those phases).

However, the icy component of the phase change material is mixed through a volumetric mixing model, which seems to provide the best fit for icy regolith mixtures especially at low to medium pore filling fractions (Siegler, et al., 2012).

The effective thermal conductivity of dry materials is expressed as the sum of thermal conductivities at each mode of heat transfer – conduction, radiation and gas conduction:

$$k_{eff} = k_{rad} + k_{cond} + k_{gas}$$

Without gas conductivity term, the effective thermal conductivity may be expressed as Watson equation (1964) modified with amorphous (or amorphous and crystalline) conductivity components:

$$k_{eff} = AT^3 + B = AT^3 + B_c k_{am}$$

The applicable models that faithfully recreate dry Lunar or Lunar analog thermal conductivities in vacuum are listed in Table 1.

Table 1. Effective thermal conductivity parameters for dry Lunar regolith in vacuum.

Lunar regolith type	Parameter A [$Wm^{-1}K^{-4}$]	Parameter B_c [-]	Amorphous conductivity, k_{am} [$Wm^{-1}K^{-1}$]	Reference
Apollo soil	$1.3 \cdot 10^{-11}$	$(9.9 \cdot 10^{-4} + 9.2 \cdot 10^{-10} \cdot \varphi^2)(1-\varphi)$	$-2.03297 \cdot 10^{-1} - 11.472 \cdot T^{-4} + 22.5793 \cdot T^{-3} - 14.3084 \cdot T^{-2} + 3.41742 \cdot T^{-1} + 0.01101 \cdot T + 2.80491 \cdot 10^{-5} \cdot T^2 + 3.35837 \cdot 10^{-8} \cdot T^3 - 1.40021 \cdot 10^{-11} \cdot T^4$	(Woods-Robinson, et al., 2019)
JSC 1-A, 1690 kgm^{-3}	$3.31 \cdot 10^{-11}$	$6.35 \cdot 10^{-4}$	$-9.53 \cdot 10^{-4} \cdot T + 2.4$	(Sakatani, et al., 2018)
JSC 1-A, 1870 kgm^{-3}	$2.39 \cdot 10^{-11}$	$9.15 \cdot 10^{-4}$		

Thermal conductivity of ice, on the other hand, is often expressed as (Hobbs, 1974):

$$k_{ice} = \frac{488.19}{T} + 0.4685 \left[\frac{W}{mK}\right]$$

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or is approximated by similar equations (Bonales, et al., 2017). Such approximations hold only to extremely low temperatures, where maximum exists (Varrot, et al., 1978). Between maximum conductivity temperature and 0 K, ice conductivity starts falling and approaches the value of 0 $Wm^{-1}K^{-1}$.

At low Knudsen number gas conductivity should be accounted for and could be calculated as (Huetter, et al., 2008):

$$k_g = \frac{k_{g,0} p \delta_p}{p\delta_p + B_g}$$

where average pore diameter and Knudsen-derived B parameter are the following:

$$\delta_p = \left[\frac{0.905}{\sqrt[3]{1-\varphi\cdot(1-F)}} - 1\right] d_p$$

$$B_g = \frac{k_B T}{\sqrt{2}\pi {d_g}^2}$$

Pressure domain

The thermal domain requires additional components from the pressure domain, which may come from a hybrid of acquired chamber data and data calculated through Darcy's law as:

$$\frac{\partial}{\partial t}(\varphi\rho_g) + \nabla\cdot(\rho_g \boldsymbol{u}) = Q_m + Q_p$$

$$\boldsymbol{u} = -\frac{\kappa}{\mu}\nabla p$$

where gas density is calculated through means of ideal gas law, vapour dynamic viscosity is assumed to be 0.017 cP and where sublimation gas source is defined as a change of bulk density of the phase change material in time subjected to mass balance constraint:

$$Q_p = \frac{\partial \rho}{\partial t}$$

$$if\ \theta = 0, \rho_{icy} = \rho_{dessicated}$$

As seen in Figure 3, gas conductivity term might result in a significant addition to the effective conductivity of porous samples up to the level of a gas conductivity in standard conditions (when water vapour is used, that counts up to around 0.02 $Wm^{-1}K^{-1}$). A different gas conductivity may be attributed to air in porous space, that is why most of those molecules have to be effectively removed during vacuum chamber pump-down.

The literature suggests that JSC-1a analogs at densities lower than 1550 kgm^{-3} should have permeability higher than 6.2 D (LaMarche, et al., 2011). An exponential fit has been created for permeability measurements used therein. The fit has also been corrected for porosity lowered by ice pore-filling fraction (i.e. effective porosity):

$$\kappa = 2\cdot 10^{-14}\cdot e^{12.469\cdot\varphi\cdot(1-F)}\ [m^2]$$

Based on permeability relations included in LaMarche et al. and FEM calculations of combined thermal and pressure domains, when vacuum chamber pressure from experimental data is applied on top of the sample (that typically took form of $p_{cham} = p_i \cdot e^{(-0.03\ldots-0.01)t}$ during pump-down), pressure evolution within the sample shows near immediate response

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(Figure 4) and equalization of pore pressure to the chamber pressure within the whole sample – which suggests that sample outgassing was successful during any pump-down operations due to very high permeability. Initial pore pressure is assumed to be $p_i = p_{cham,i}(1 - F)$, while initial chamber pressure was typically $p_{cham,i} \approx 600 \ldots 650\ [Torr]$. Thus, pore-space gas conductivity should come solely from chamber pressure and sublimation-based mass movement. Nevertheless, two other scenarios have been analysed – when the given permeability is 100 times lower and when it is 1000 times lower. The results can be found in Figure 5.

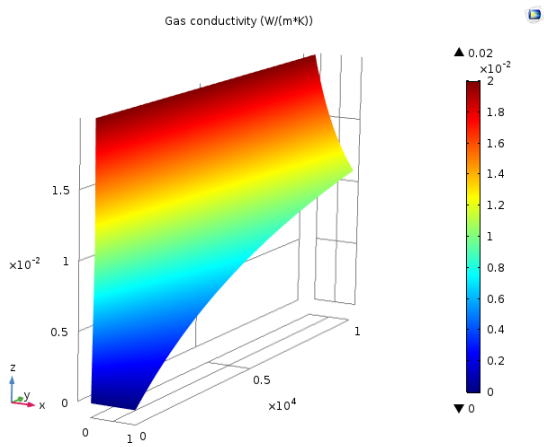

Figure 3. Gas conductivity model variable with temperature (x, K, 10^3) and pressure (y, Pa).

As seen below, permeability highly influences the heating process. Unsuccessful pressure decrease in regolith pores might influence not only the heating process itself but also it could switch phase change from sublimation-based to fusion-based, at least locally where pressures and temperatures are above water triple point.

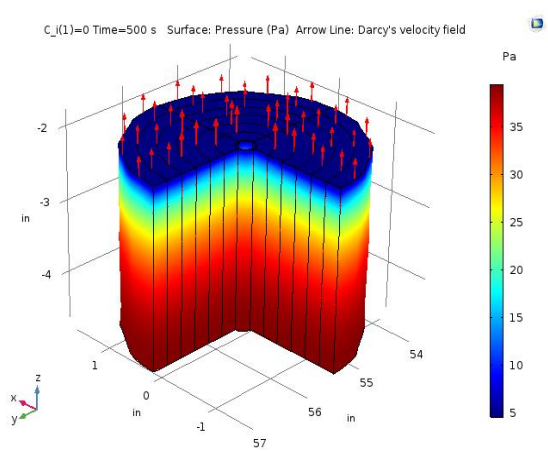

Figure 4. Pore-pressure evolution during pump-down operations in a dry JSC-1A sample.

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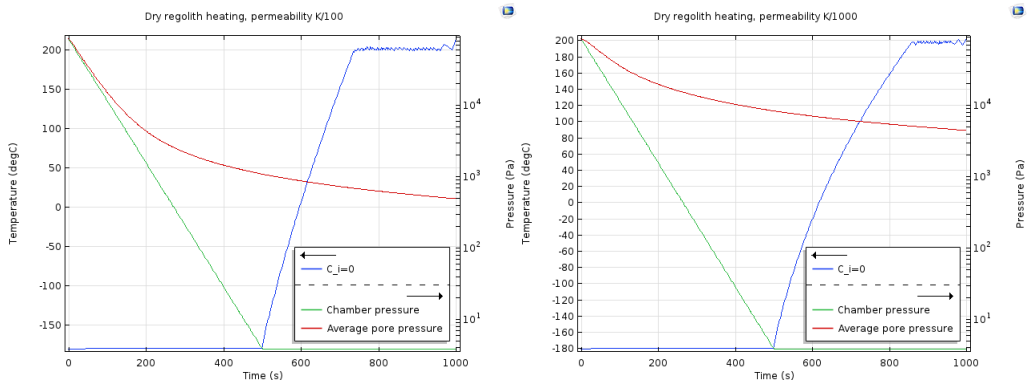

Figure 5. Pore pressure and heater temperature evolution in dry regolith samples when permeability is lower than anticipated.

Icy Lunar regolith

When effective thermal conductivities of Lunar regolith and JSC-1A analog are calculated based on previously mentioned theoretical models, a discrepancy between those two materials is visible especially at higher densities of dry regolith – mainly due to different amorphous conductivity approximations. Nevertheless, those materials express strong similarity when ice is present because ice conductivity starts becoming dominant even at very low ice filling fractions. An influence of porosity and ice content on thermal conductivities of Lunar regolith in vacuum is presented in an example in Figure 7.

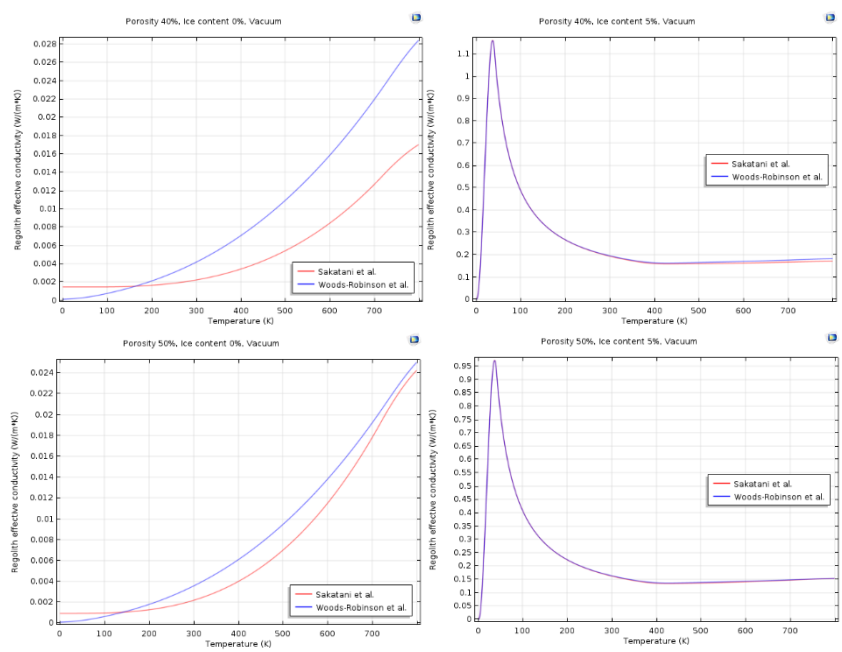

Figure 7. Theoretical models of effective thermal conductivity of dry and icy Lunar regolith in vacuum. The effects of porosity change and average ice content as in LCROSS plume (Colaprete, et al., 2010) are presented.

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With regard to heat capacity, multiple models exist that reflect materials used within this study. Their approximations are presented within Table 2. Their exemplary calculation is presented in Figure 7.

Table 2. Ice and dry basalt heat capacity models.

Material type	Heat capacity equation $\left[\frac{J}{kgK}\right]$	Reference
Apollo fit	$C_{p,dry} = -5.84 \cdot 10^4 + 3.64 \cdot 10^3 T - 2.03T^2 - 4.14 \cdot 10^{-3}T^3 + 3.87 \cdot 10^{-6}T^4 \; for \; 200K < T < 400K$ $C_{p,dry} = -1.12 \cdot 10^{-1}T + 9.62 \cdot 10^{-2}T^2 - 1.26 \cdot 10^{-3}T^3 + 7.69 \cdot 10^{-4}T^4 - 2.37 \cdot 10^{-8}T^5 + 2.84 \cdot 10^{-11}T^6 \; for \; 10K < T < 200K$	(Woods-Robinson, et al., 2019)
ice	$C_{p,ice} = 7.73 \cdot 10^{-3}\left(1 - e^{-1.263 \cdot 10^{-3}T^2}\right) \cdot \left(1 + e^{-3\sqrt{T}} \cdot 8.47 \cdot 10^{-3}T^6 + 2.0825 \cdot 10^{-7}T^4 e^{-4.97 \cdot 10^{-2}T}\right)\left[\frac{J}{gK}\right]$	(Shulman, 2004)

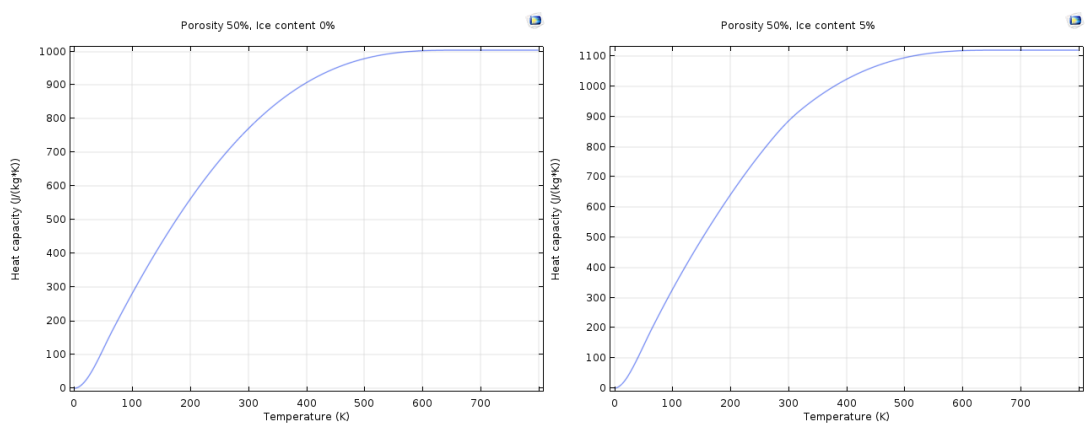

Figure 7. Lunar regolith heat capacity calculation example. Left - for dry regolith with 50% porosity, right - for 5% icy regolith with 50% porosity. Phase change enthalpy was not accounted for in these plots.

The model running on the set of inputs mentioned above is applied to a simplified CAD model of the vacuum chamber used in experiments. The geometries within the chamber had been measured, recreated and respective materials were assigned to them. Mesh that has been selected and built for the chamber focuses on the phase change material, where the finest mesh element size parameters have been assigned (as seen in Figure 8).

The approach used within this study is based on iterative computation of the heat transfer model and comparative analysis of theoretical results with experimental data. The goal of the comparative analysis is refinement of effective thermal conductivity parameters that results in the best fit between modelled and experimental data. The modelled and experimental data is analysed over the first 1,000 seconds from the start of heating. The 3D visualization of modelled heat transfer within regolith samples may be found in Figure 9.

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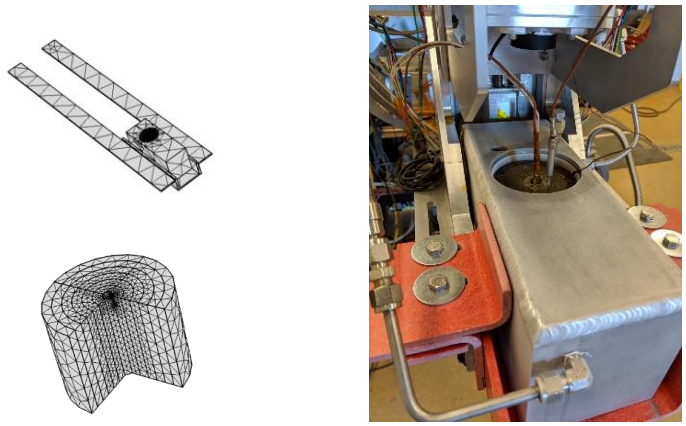

Figure 8. Mesh model of the chamber interior (left), cross-section of the sample (middle) and picture of experimental situation (right).

Discrepancies between theoretical and experimental data allow to understand the dynamics of icy regolith heating process but most of all they point out, which effective parameter could to be changed to recreate the process with the model.

Whenever the effective thermal conductivity is calculated as mentioned above – the heater temperature results do not fully match experimental data. Over the course of the study it has been determined that two parameters have the biggest impact on discrepancies between modelled and experimental data series:

- dry regolith thermal conductivity – the lower it gets, the heater shows faster temperature increase rate and higher inertia; low thermal conductivity of dry samples does not match experimental data, which may be a result of very fine regolith used in the study or gas conductance higher than anticipated (which on the other hand may come from lower permeability than used in this study)

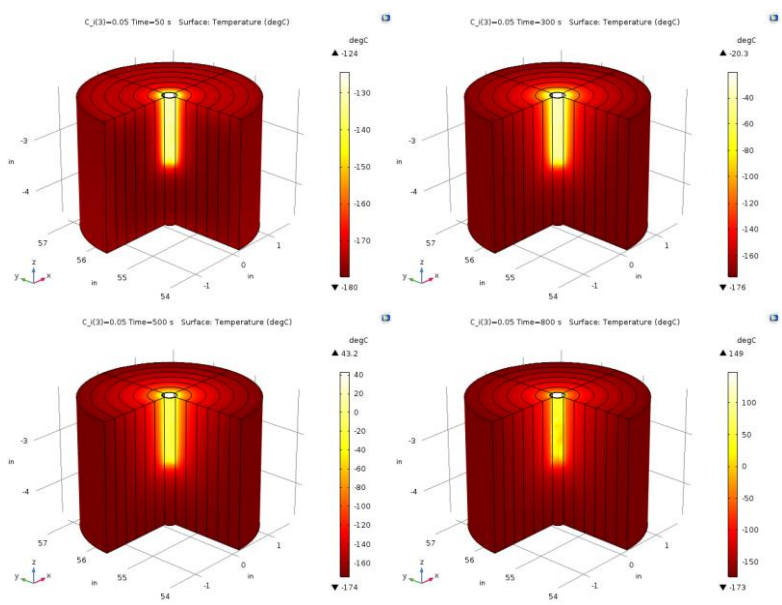

Figure 9. Icy regolith sample temperature evolution modelling over 800 seconds. In this example, a 5% wt icy Lunar regolith is being used.

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- phase change transition temperature and phase change transition interval – during various runs it has been determined that phase change transition temperature starts occurring between -20 and 20 ^{0}C, with rapid sublimation (seen as pressure spikes or even sample explosions) occurring around 0 ^{0}C. However, low phase change transition intervals (orders of couple of Kelvins) result in large increase of heater temperature growth rate around the transition temperature – this does not match experimental data, which suggests that transition interval is large and transition is much smoother. That is why the calculations solve phase change for transition temperature of 0 ^{0}C (peak of phase change enthalpy increase), and transition interval between -50 and 50 ^{0}C (phase change enthalpy is proportionally spread over that temperature range). However, both the transition temperature and transition interval ought to be considered as rough estimates. The model does not take into account hysteresis effects as they could not be inferred from the data but should be considered in future work.

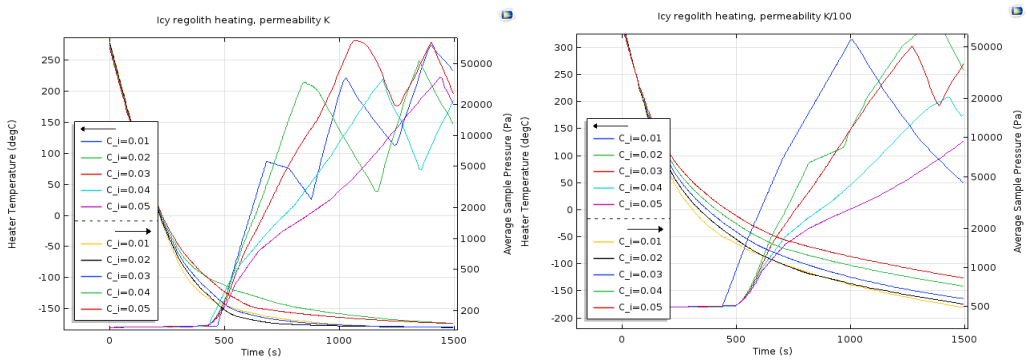

Figure 10. Heater temperature and pressure evolution in experimental situation of the Model with anticipated permeability (left) and 100 times lower permeability (right).

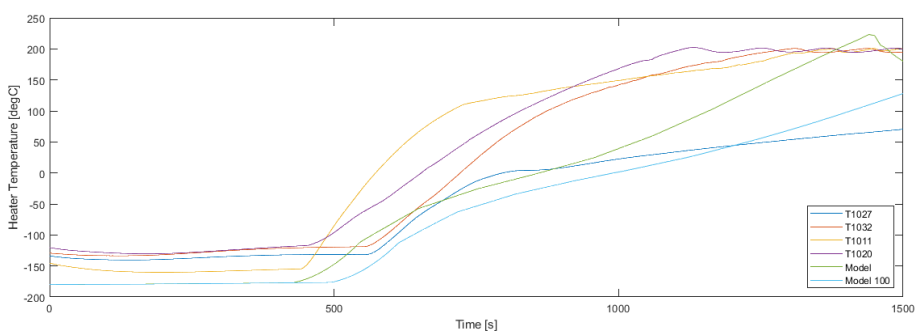

Figure 11. 5% icy regolith heating experimental data (T*) with regard to the model with values as in Figure 11. In the experiments, the target 200 ^{0}C is reached within 600 s (T1032), 650 s (T1020), 930 s (T1011) or even 3000 s (T1027).**

All the runs suggest that the dry regolith thermal conductivity should be increased in the model to provide a better fit for both dry and icy regolith heating tests, as neither higher pressure nor lower permeability account for that difference fully. This is especially visible with dry regolith heating experiments as well as with temperature evolution during and after phase change transition. However, it only means that the regolith used in the experiments had been validated to

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hold slightly different material properties from Lunar and Lunar analog materials, while properties of the latter are considered right.

As seen in the exemplary juxtaposition of tests and models results (Figure 11), the runs are highly variable even if the same methodology and material is used. The runs also suggest that the investigated processes are stochastic, which is visible in the models and experiments – small change of initial parameters results in different heating curve.

Even though the experimental process went through the same steps of preparation and execution and results are highly variable, the expected trend in heating versus ice content is maintained (Figure 12) – low ice content surfaces tend to be easier to heat to a specific temperature, while high ice content surfaces are very hard to heat up.

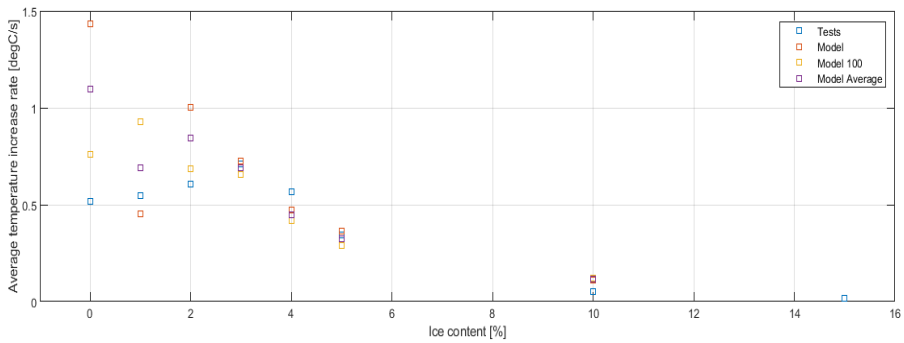

Figure 12. Average temperature increase rates at specific ice content regoliths. Rates are calculated as slopes of linear regression of heater temperature evolution between start of heating and time of maximum temperature reach (at target 200 ^{0}C or below at highest after more than 10,000 s).

CONCLUSIONS

Heat and mass transfer within icy planetary deposits is an extremely computationally-intensive process. Nevertheless, it may be studied using approaches like FEM modellings. This study aimed to validate and iterate thermal properties of icy deposits based on experimental data. It proved that the studied process parameters and variables are highly interdependent but when modelled physics is thoroughly considered, thermal evolution trends may be recreated.

The study proved that our understanding of thermal properties of dry materials and their volumetric mixtures with ice is proper only if dynamism of ice phase change and pressure domain are taken into account.

Co-dependent thermal, pressure and mass evolution render the process of heat transfer and water extraction in planetary environments counter-intuitive and non-linear. This is the area of further research based on the provided model.

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NOTATION

α thermal diffusivity, $\frac{m^2}{s}$
α_m phase change mass fraction, dimensionless or %
β gas specific constant, dimensionless
A radiative parameter of conductivity, $\frac{W}{mK^4}$
B_g factor, $\frac{J}{m^2}$
B_c solid parameter of conductivity, dimensionless
C_i ice content, % wt
C_r regolith content, % wt
C_p specific heat of phase change material, $\frac{J}{kgK}$
$C_{p,x}$ specific heat of x phase or material, $\frac{J}{kgK}$
d_g diameter of a gas molecule, Å or m
d_p diameter of a particle, μm or m
δ_p pore size estimation, μm or m
F pore filling fraction, dimensionless or %
k_B Boltzmann constant, $\frac{J}{K}$
k thermal conductivity of phase change material, $\frac{W}{mK}$
k_x thermal conductivity of x phase or material, $\frac{W}{mK}$
$k_{g,0}$ thermal conductivity of a gas in standard conditions, $\frac{W}{mK}$
κ permeability, D or m^2
L latent heat of icy regolith phase change, $\frac{kJ}{kg}$
L_s latent heat of water ice sublimation, $\frac{kJ}{kg}$
λ mean free path of gas molecules, m
M_s mass of a sample, kg
μ fluid viscosity, cP or $Pa \cdot s$
p pressure, $Torr$ or Pa
φ porosity, dimensionless or %
Q heat source, W
$Q_{m,p}$ gas source − sink rate, $\frac{kg}{m^3 s}$
q heat flux at unit length, $\frac{W}{m}$
ρ density of phase change material, $\frac{kg}{m^3}$

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$\boldsymbol{\rho_x\ density\ of\ x\ phase\ or\ material, \frac{kg}{m^3}}$
$\boldsymbol{T\ temperature,}$ ℃ $\boldsymbol{or\ K}$
$\boldsymbol{t\ time, s}$
$\boldsymbol{\theta\ icy\ phase\ fraction, dimensionless\ or}$ %
$\boldsymbol{u\ velocity\ field, gas\ flow, \frac{m}{s}}$
$\boldsymbol{V_s\ volume\ of\ a\ sample, m^3}$

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